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Korea’s First Full-Stack Humanoid Clears Backflip: IGRIS-C Is Already on Factory Floors

Robros built IGRIS-C’s motors, control stack, and AI in Seoul; IGRIS-C is already on logistics floors

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Published: Jul 31 2026, 4:40 PM EDT





(PHOTO: ROBROS.CO.KR)

On July 29, 2026, a Seoul-based startup most people outside Korea had never heard of published a short video that circulated immediately among humanoid robotics engineers worldwide. In it, a bipedal robot performed a windmill spin, a backflip, a series of kicks, and a squat sequence. The robot is IGRIS-C, built by Robros. The company claims it is the first bipedal humanoid independently developed by a domestic Korean company to achieve this level of [dynamic full-body motion](#). What engineers noticed was not just the movement — it was what the movement was built from: hardware, control algorithms, and AI all developed in Seoul, without foreign actuators, off-the-shelf control frameworks, or external motion capture systems.

That distinction mattered more than it might have a week earlier. On July 28, 2026 — one day before the Robros demo circulated — the Federal Communications Commission added all foreign-produced humanoid and quadruped robots to its national security [Covered List, barring new Chinese-made models from US import, marketing, and sale](#). Chinese manufacturers hold roughly 85 percent of global humanoid market volume. The market gap that just opened does not fill itself with American machines — at least not at current price points. Korea's full-stack independent humanoid showing up the following morning is not coincidental timing. It is a credential for a specific vacancy.

Robros is not a conglomerate, a government research lab, or a spinout from a defense program. It is a startup founded in 2020 with a team of over 40 people, backed by \$10.6 million in total venture funding, that has built its own hardware and software stack without the manufacturing scale of Samsung or LG or the institutional resources of KAIST. That is the specific context in which the July 29 video carries weight.

Read more: [China Humanoid Patents Surge Fivefold in Key AI Category as FCC Bans New Imports](#)

Why a Backflip Is an Engineering Benchmark, Not Entertainment

It is worth being precise about what the Robros demonstration actually verified. A backflip is not a useful industrial motion. It is a stress test for a specific engineering capability called dynamic whole-body control.

balance, and joint dynamics across all available degrees of freedom, as [documented in the whole-body control literature](#). Conventional robots accomplish tasks from a stationary base, with upper-body motion managed independently of lower-body stability. That architecture fails when the robot needs to move and work at the same time — carry a load across a warehouse floor, or adjust its grip while walking.

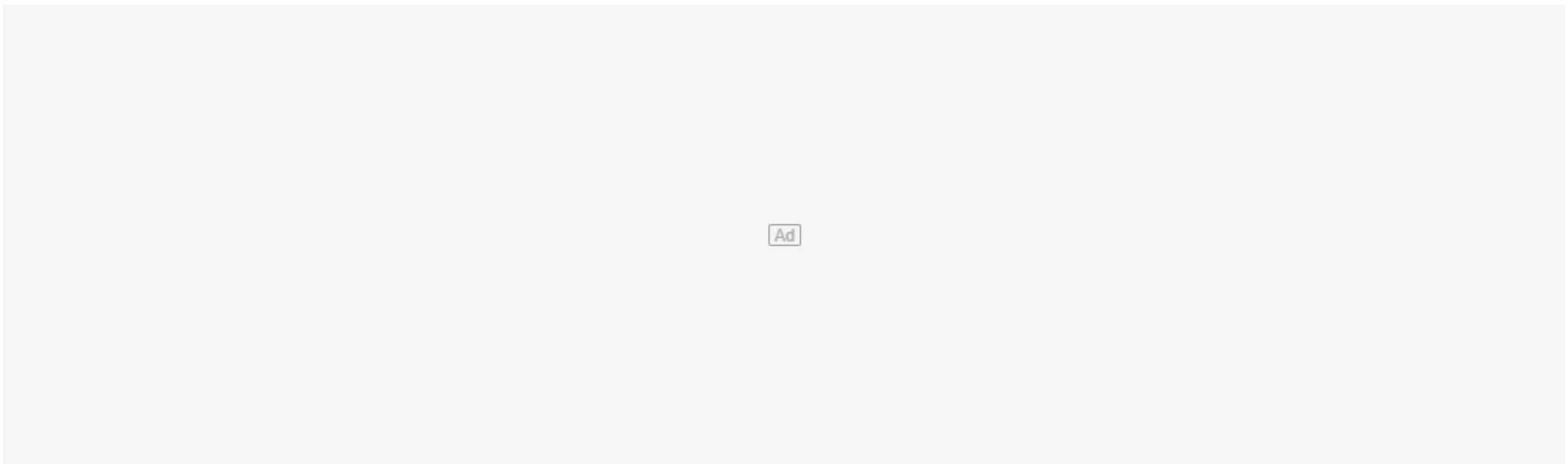


The backflip is the maximum-stress test for WBC because it demands instantaneous, large-amplitude changes in the robot's center of gravity while the system is airborne and rotating. Every joint in the body must coordinate in real time to stick the landing, as [Robros explains in its technical description of the demonstration](#). If the control stack handles that, it can handle the less dramatic but more commercially relevant version of the same problem: a robot that walks and carries at the same time, or that maintains balance while both arms are applying force.

This is why dynamic motion demonstrations are taken seriously by engineers even when the movement itself has no practical use. The question being answered is: does this control architecture produce the real-time full-body coordination that industrial tasks require?

How IGRIS-C's Control Stack Actually Works

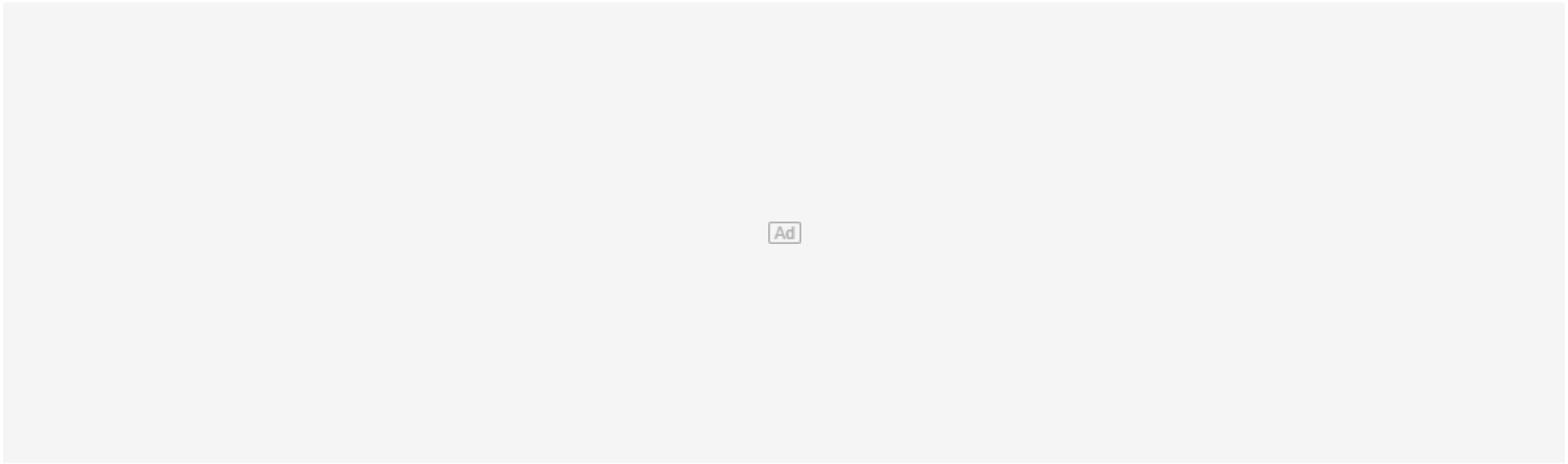
IGRIS-C's ability to perform a backflip is a product of two design decisions made before the robot was built: the choice of actuation architecture and the choice of learning methodology.



On the hardware side, Robros uses a [low reduction ratio drive system with torque-based control](#). This is a specific architectural choice with a specific tradeoff. Higher gear reduction ratios (50:1 to 100:1) found in typical collaborative robots make joints safer and more controllable at slow speeds — but they reduce the actuator's ability to respond quickly to disturbances. IGRIS-C's lower reduction ratio (reported at 30:1 per Aparobot) delivers what Robros calls "high torque transmission," with a [maximum joint torque of 150 newton-meters](#). The tradeoff: the robot responds faster to control commands and absorbs physical shocks more readily — which is what a backflip requires — but the architecture demands a more sophisticated control system to maintain stability at rest.

[with policies then transferred to the physical hardware \(the sim-to-real approach\).](#) [Imitation learning \(IL\) handles manipulation tasks](#): human operators drive the robot via a "master hand" interface, collecting demonstrations that train the robot to reproduce the same task in similar environments. The combination — RL for locomotion, IL for manipulation — is now standard among advanced humanoid developers. It is the architecture that allows the same platform to walk dynamically and handle objects precisely, without requiring one subsystem to solve both problems.

The hands themselves reflect the same design logic. Each hand uses a [tendon-driven structure — actuators and joints connected by wires — delivering six active degrees of freedom per hand and a total of 11 including passive degrees of freedom](#). The result: each hand weighs approximately 300 grams (10.6 oz) with a per-hand payload capacity of 3 kilograms (6.6 lbs). Lightweight hands on a humanoid matter because the robot's arms act as long lever arms — excess mass at the end of the arm increases the torque the shoulder joint must manage. The tendon-driven architecture trades some mechanical complexity for substantially reduced distal mass, which simplifies the stability problem for the whole-body controller.

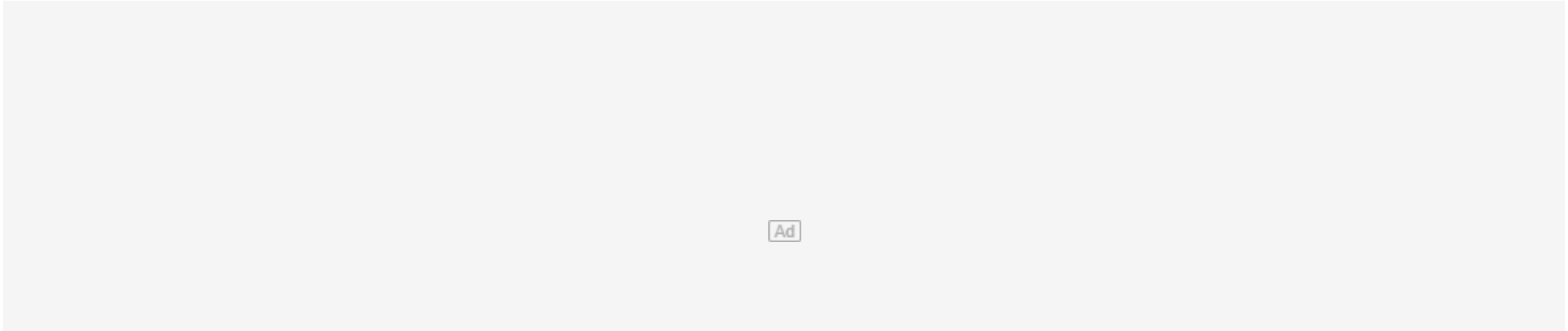


IGRIS-C's onboard compute is a pairing of an NVIDIA Jetson Orin and an Intel Core i7, supported by a stereo RGB camera array and a rear-facing depth sensor, per [Robros' published hardware specifications](#). The robot stands 154 centimeters (5 feet 1 inch) tall and weighs 56 kilograms (124 lbs), with a dual-arm lift capacity of 6 kilograms (13.2 lbs) — specifications oriented toward industrial utility rather than athletic performance.

Korea's Independent Stack vs. the Competition

What makes the Robros demonstration specifically notable in the competitive landscape is the [full-stack independence claim: hardware design, actuator manufacturing, control algorithms, and AI software all developed in-house](#), without reliance on foreign components or licensed control frameworks.

This claim distinguishes IGRIS-C from a significant portion of the global humanoid field, including many US and European startups that build their robots using Chinese actuators (often Unitree or affiliated suppliers), off-the-shelf compute modules, and open-source control frameworks from ROS (Robot Operating System) or research institutions. A robot assembled from globally sourced components is fast and cost-efficient to develop. It is also dependent on the supply chains for those components.



The frame of comparison that Robros invites is with Boston Dynamics' Atlas and with Chinese systems. Boston Dynamics has long been the reference for what a humanoid can do physically; the RAI Institute, led by Boston Dynamics founder Marc Raibert, has described its platform as producing gymnastic maneuvers and natural walking gait from the same learning framework — a unified whole-body approach that [allows control policies trained in simulation to run on physical hardware without additional tuning](#). The key point: the same underlying learning architecture produces both a backflip and a walking gait. IGRIS-C's claim is to have reached that same architectural capability through an entirely domestic engineering effort.

On the Chinese side, the reference event is Unitree Robotics' performance at China's 2026 Spring Festival Gala on February 17, 2026, where G1 and H2 machines executed complex martial arts routines including [the world's first continuous freestyle table-vaulting parkour, an aerial flip with a maximum height approaching approximately 10 feet \(3 meters\), and a two-step wall-assisted backflip](#). Those demonstrations established the benchmark that any serious humanoid must now clear to be taken seriously on dynamic motion.

Robros cleared it six months later, with hardware built in Seoul.

Where IGRIS-C Actually Is Right Now

The demo is not the story. The floor deployments are.

Robros is currently conducting [proof-of-concept evaluations at logistics companies including Lotte Global Logistics and at shipyard sites](#) — confirming industrial applicability beyond laboratory conditions. IGRIS-C units have been [supplied to Kwangwoon University, Kyung Hee University, and Sogang University](#) for educational and research applications. The company has been [selected for multiple projects under a humanoid demonstration program run by the Korea Institute for Robot Industry Advancement \(KIRIA\)](#).

The company's positioning — reliability at work sites, not speed records — is legible in the specifications. IGRIS-C tops out at 3.6 kilometers per hour (2.2 mph), slower than Unitree and Figure's machines, by the company's own acknowledgment. The architecture prioritizes the torque response and stability characteristics that industrial applications require over the speed and agility that demo videos reward.



Korean media has also reported that IGRIS-C robots are being piloted inside LG facilities, though LG has not publicly confirmed this.

Where Robros Fits in Korea's Humanoid Ecosystem

Robros is not the only Korean humanoid story, but it is a different kind of story than the ones already on record.

Shallow-π VLA model already tested in factory settings. LG's CEO-level robotics business center, [backed by 45 million motors per year of appliance-motor manufacturing experience](#), is building its AXIUM actuator brand to supply both its own CLOi D humanoid and external robot makers. KAIST's Pibot, meanwhile, [demonstrated naval helmsman duties at the ROK Navy's Changwon training facility on July 24, 2026](#), as part of South Korea's "Navy Sea GHOST" doctrine responding to a demographic conscription crisis.



Robros occupies a different tier: a startup founded in 2020, now with a team of over 40 people, that has built its own hardware and software stack without conglomerate backing or government laboratory resources. As of June 2026, it had raised \$10.6 million from investors including HB Investment, KB Investment, Lotte Ventures, and Mirae Asset Securities, per PitchBook. At the NextRise 2026 startup fair co-hosted by the Korea Development Bank and the Korea International Trade Association, held June 18-19, 2026 at COEX in Seoul, [Robros won the Next Innovator Prize in humanoid robotics — the only robotics company to do so among that year's 32 honorees](#).

What IGRIS-C Cannot Do Yet — and What That Tells Researchers

The industrial pitch works only if IGRIS-C can handle the hard parts of industrial work, not just the dramatic parts.

Alicia Veneziani, Global VP at Singapore-based manipulation robotics company [Sharpa](#), put the critical challenge directly at CES 2026: even as robots increasingly master dancing and backflips, she argued, manipulation remains the real bottleneck for useful, autonomous robots. The distinction she was drawing is between dynamic locomotion — now cleared by a growing number of platforms — and reliable dexterous manipulation in variable, unstructured environments. A backflip proves the control architecture can handle maximum center-of-gravity disruption. It does not prove the robot can grip an irregularly shaped object reliably for 1,000 consecutive cycles in a live warehouse.



IGRIS-C's industrial PoC evaluations at Lotte Global Logistics and at shipyard sites are the tests that actually answer that question. What Robros has not yet published — and what any serious procurement evaluation will require — is structured performance data from those deployments: task completion rates, failure modes, error recovery behavior, and

Read more: [Humanoid Robot Steers Naval Vessel for First Time as South Korea Faces Crew Crisis](#)

Funding and What Comes Next

Korean humanoid valuations are rising sharply as the global market moves from laboratory demonstrations to early commercial deployments. Robros, with its demonstrated dynamic capability and active industrial PoC work, is among the companies expected to seek further funding at significantly higher valuations than its current \$10.6 million total raised, per PitchBook. The timing of the July 29 demo — a clear capability statement following the NextRise innovation prize and the KIRIA program selections — signals deliberate positioning for an upcoming fundraiser.

The broader question is not whether IGRIS-C can backflip. That question was answered July 29. The question is whether the same control architecture, the same full-stack independence, and the same industrial focus translate into deployment performance at the scale that logistics and manufacturing partners require. Boston Dynamics spent years proving Atlas could do gymnastic maneuvers before electric Atlas became a commercial industrial product. The gap between the demo and the deployment is where most humanoid startups stall.



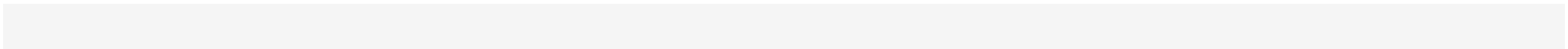
Robros' specific argument is that the deployment work started before the demo — that the factory-floor and logistics evaluations are not something they plan to do because the video went viral, but something they were already doing before anyone outside Korea had heard of them. That claim, if the PoC data supports it, changes the competitive calculus. A startup that can backflip and has been quietly building industrial track record at the same time is a different kind of competitor than one that has mastered the demo circuit.

As of July 31, 2026, a small company from Seoul has earned a place on the shortlist that engineers are assembling for the post-FCC-ban market. Whether it stays there depends on what the Lotte Global Logistics and shipyard data shows.

Frequently Asked Questions

What is dynamic whole-body control, and why does a backflip test it?

Dynamic whole-body control (WBC) is a robotics architecture that treats the entire robot as a single coupled system — managing locomotion, balance, and manipulation simultaneously rather than with separate controllers for walking and working. A backflip tests it at maximum stress because it requires instantaneous, large-scale changes in the robot's center of gravity while the system is airborne and unsupported, demanding that every joint in the body coordinate in real time. If the WBC architecture handles a backflip, it has demonstrated the same capability — real-time full-body coordination under dynamic disturbance — that industrial tasks like walking while carrying a load or manipulating an object while in motion require.



How is Robros' IGRIS-C different from Chinese humanoid robots like Unitree?

The core difference is the sourcing of the full technology stack. Unitree's machines use components and control frameworks that include both in-house and supplier-sourced elements; the company's strength is manufacturing scale and cost efficiency. IGRIS-C is claimed to be built from hardware, control algorithms, and AI software all developed internally in Seoul, with no foreign actuators or licensed control frameworks. The robots are also positioned differently: Unitree's G1 and H2 excel at speed and acrobatic performance (the Spring Festival Gala demonstrations in February 2026 are the reference point), while IGRIS-C is slower (2.2 mph versus higher speeds for competing platforms) but positioned for reliability in industrial settings rather than demo performance. After the FCC Covered List designation on July 28, 2026 barred new Chinese humanoid models from US markets, Robros' non-Chinese origin and full-stack independence became a market eligibility credential, not just a technical claim.

Is IGRIS-C available commercially, and what does it cost?

Robros opened preorders for IGRIS-C in mid-2025 and began supplying units to universities (Kwangwoon, Kyung Hee, Sogang) for educational and research use. The robot is currently positioned as a research and industrial PoC platform rather than a mass-market commercial product. Pricing has not been publicly disclosed in confirmed sources available as of this article's publication. The company's funding (\$10.6 million total raised, per PitchBook) and team size (over 40 people) suggest it remains in the scale-up phase before broad commercial availability.

What does it mean that Robros is "full-stack independent" — and why does that matter after the FCC ban?

Full-stack independence means Robros designed and manufactured its own hardware (frames, joints, actuators), developed its own control algorithms (the software that translates desired motion into joint commands), and built its own AI models (for walking and manipulation), without sourcing these from foreign suppliers or licensing them from third parties. It matters specifically after the FCC Covered List action of July 28, 2026 because that designation barred new Chinese humanoid robots from US markets. Chinese manufacturers hold roughly 85 percent of global humanoid market volume, and their hardware is deeply embedded in the supply chains of many non-Chinese robotics companies. A platform built entirely from domestic components — with no Chinese actuators, no Chinese control software, and no Chinese IP in the stack — is not exposed to the supply chain implications of those restrictions. That makes Robros' architecture a specific eligibility credential for US and allied procurement that cannot be matched by robots assembled from globally sourced parts.

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1.

A humanoid robot, possibly a Shadow robot, is shown in a grassy field. It has a dark, metallic body with a head that looks like a camera or sensor. It is standing on two legs and appears to be in motion.

[Ukraine Sets Six-Month Humanoid Kill Benchmark With Robot That Cannot Work in Rain](#)
2.

A close-up of the Samsung logo on a yellow background. The logo is in black and is slightly tilted. The background is a bright yellow, possibly a light source or a screen.

[Galaxy Tab S12 Ultra Gains Speed, Loses Six Hours of Battery Before Oct. 7 Launch](#)
3.

A close-up of a hand wearing a ring. The ring is silver and has a small, round, clear stone. The hand is holding a small object, possibly a ring or a piece of jewelry.

[Oura IPO Pulled Hours Before Pricing: More Than Market Jitters Held Investors Back](#)
4.

The World Labs logo, which features a globe with a fish inside it. The text "World Labs" is written across the globe. The background is blue.

[AMD Buys World Labs: Fei-Fei Li Will Now Shape the Chips That Run Physical AI](#)
5.

A group of people standing behind a podium at a press conference. There are several people, some in suits and some in military uniforms. They are all looking towards the camera. The background is a blue wall with a logo.

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